

MAAT–SO(10) Extended Benchmark Framework

From Phenomenological Bridge to Response-Closed Multi-Scale GUT Selection

Christof Krieg

MAAT Project – Independent Research

<https://maat-research.com>

2026

Abstract

The first MAAT–SO(10) phenomenological bridge paper introduced a structural selection layer over SO(10)-motivated gauge and Yukawa benchmarks. Its main result was deliberately limited: a one-loop gauge diagnostic located a plausible GUT-scale region but failed precision unification, while a third-generation b - τ Yukawa benchmark produced a highly coherent structural solution with a modest effective bottom-threshold correction.

This note turns those limitations into a concrete extended benchmark framework. The goal is not to claim a complete SO(10) construction, a precision GUT fit, a derivation of the Standard Model spectrum, or a solution of vacuum selection. Instead, it specifies the next reproducible MAAT–SO(10) layer: two-loop and threshold-aware gauge diagnostics, expanded Yukawa and neutrino tests, response-closed structural weights, v1.2.1-compatible emergent robustness, multi-scale defect-field selection, RG generator-stationarity, CCI-style regime diagnostics, and explicit falsification baselines.

The extended framework reframes MAAT–SO(10) from a single illustrative bridge into a versioned testing programme. A successful future implementation must beat gauge-only, threshold-only, complexity-only, shuffled-defect, and random-weight baselines under predeclared metrics. A failure to do so would be informative: it would show that the present structural supports do not add predictive information beyond conventional GUT diagnostics.

1 Purpose and Status

SO(10)-motivated grand unification provides a natural dynamical base for testing structural selection: gauge unification, Yukawa relations, threshold corrections, neutrino masses, proton decay, hierarchy pressure, and vacuum choice all create a high-dimensional constraint landscape.

The existing MAAT–SO(10) benchmark should therefore be read as a first bridge, not as a finished theory. Its strongest use is diagnostic: it tells us which parts of the construction are already structurally coherent and which parts require a harder GUT-level implementation.

Scope. This paper is an extended benchmark specification. It does not introduce new observational evidence and does not modify the underlying SO(10) dynamics. It defines how the next MAAT–SO(10) test should be built, what should count as success, and what should count as failure.

2 Short MAAT Recap for New Readers

The MAAT layer used here is a structural ranking layer, not a replacement for the underlying field theory. The dynamical base determines which configurations or parameter regions are

admissible. MAAT assigns additional diagnostic coordinates to those admissible candidates and asks which ones are structurally coherent, balanced, dynamically active in a controlled way, and internally connected.

The four primary supports are:

- H : equation or consistency support. In a GUT benchmark this measures how well a candidate satisfies the chosen dynamical and matching equations.
- B : balance or constraint support. In $\text{SO}(10)$ this includes conservation, matching, threshold, and stationarity consistency.
- S : controlled activity support. This measures whether the candidate has enough dynamical structure to be nontrivial without becoming unstable or overfit.
- V : connectivity support. This measures whether gauge, Yukawa, threshold, neutrino, and vacuum sectors form a coherent coupled structure rather than disconnected fits.

Each support is obtained from a non-negative defect,

$$\Gamma_a = \frac{1}{1 + d_a}, \quad d_a \geq 0, \quad a \in \{H, B, S, V\}.$$

Thus low defect means high support. The compact structural score has the form

$$F_{\text{MAAT}} = - \sum_{a \in \{H, B, S, V\}} \lambda_a \log(\epsilon + \Gamma_a) - \lambda_{\text{cl}} \log(\epsilon + R_{\text{rob}}),$$

where λ_a are not headline free parameters in the extended benchmark. They are estimated by the response-closure protocol described below.

Robustness is derived rather than introduced as a fifth primitive sector:

$$R_{\text{resp}} = (HBV)^{1/3}, \quad R_{\text{rob}} = \min \left[(HBV)^{1/3}, (HBSV)^{1/4} \right].$$

This distinction is important. A candidate can satisfy a single constraint well and still fail structurally if the supports do not remain coherent under the full set of GUT stresses. In the present paper, MAAT should therefore be read as a falsifiable candidate-ranking language over $\text{SO}(10)$ -motivated benchmarks, not as an independent GUT model.

3 Current Benchmark Baseline

The current repository implementation contains two benchmark scripts:

- `maat_so10_gauge_2d_fit.py`;
- `maat_so10_yukawa_coupled_rg_v04.py`.

The first script performs a one-loop gauge-sector diagnostic without threshold corrections. The second tests an $\text{SO}(10)$ -motivated b - τ high-scale boundary condition with coupled one-loop third-generation RG flow and an effective bottom threshold.

Quantity	Gauge diagnostic	Yukawa diagnostic
Scale	$M_{\text{GUT}} \simeq 6.64 \times 10^{15} \text{ GeV}$	$M_{\text{GUT}} \simeq 1.86 \times 10^{16} \text{ GeV}$
Main residual	$\chi_{\text{SM}}^2 \simeq 100.79$	$\chi_{\text{Yukawa}}^2 \simeq 2.43 \times 10^{-4}$
Threshold	none	$\Delta_b \simeq 0.0506$
H	0.9962	0.9999999
B	0.6194	0.9999997
S	0.6796	0.9999999
V	0.3554	0.9975
R_{rob}	0.6030	0.9992
Status	plausible scale, no precision unification	coherent b - τ compatibility benchmark

Table 1: Current MAAT–SO(10) benchmark baseline. The contrast is the key motivation for the extended framework: gauge unification requires a harder threshold- and spectrum-aware treatment, while the third-generation Yukawa benchmark already shows a structurally coherent compatibility region.

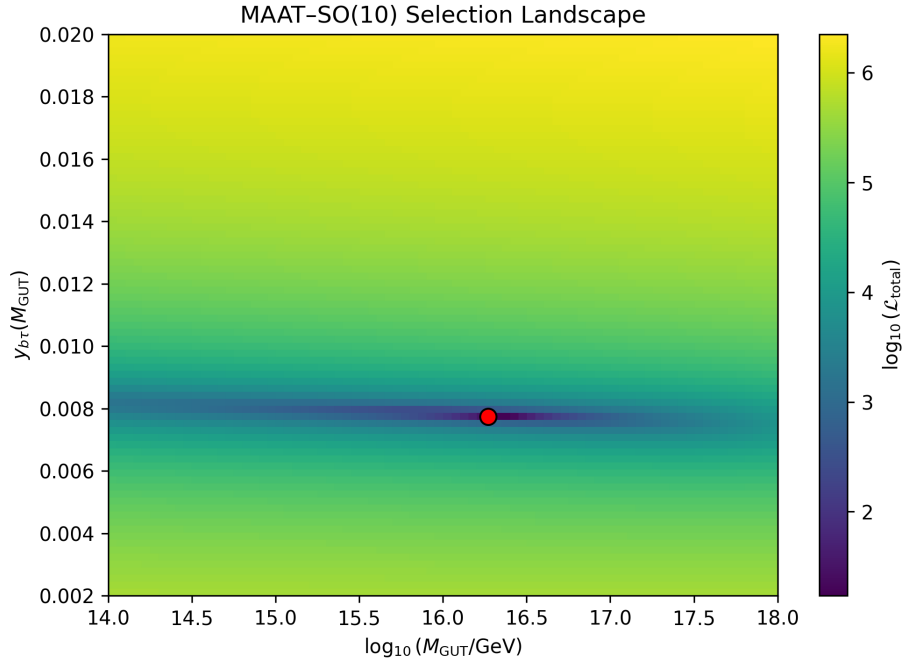


Figure 1: Existing MAAT–SO(10) Yukawa selection landscape. The important feature is not a unique point prediction, but a narrow structural valley in the $(\log_{10} M_{\text{GUT}}, y_{b\tau})$ plane. The extended framework should test whether such valleys persist under additional constraints.

4 v1.2.1 Closure: Four Primary Supports

The extended SO(10) framework must remain consistent with MAAT v1.2.1. The primary supports are

$$H, \quad B, \quad S, \quad V.$$

Robustness is not a fifth freely fitted primitive sector. It is derived:

$$R_{\text{resp}} = (HBV)^{1/3}, \tag{1}$$

$$R_{\text{rob}} = \min \left[(HBV)^{1/3}, (HBSV)^{1/4} \right]. \tag{2}$$

The stability of a candidate is then

$$\text{Stability} = R_{\text{rob}}.$$

This matters for SO(10), because otherwise a fifth independent robustness weight can hide gauge-threshold stress or proton-decay pressure by double counting correlations already present in H , B , and V .

5 Extended SO(10) Defect Vector

The first benchmark used compact gauge and Yukawa defects. The extended benchmark should expose the physical stress channels explicitly. A minimal defect vector is

$$d = (d_{\text{eq}}, d_{\text{cons}}, d_{\text{dyn}}, d_{\text{conn}}, d_{\text{gauge}}, d_{\text{Yukawa}}, d_{\text{threshold}}, d_{\text{proton}}, d_{\nu}, d_{\text{hierarchy}}, d_{\text{complexity}}).$$

The domain-specific components should include:

$$d_{\text{gauge}} = \chi_{\text{gauge}}^2(\alpha_1, \alpha_2, \alpha_3), \quad (3)$$

$$d_{\text{threshold}} = \sum_i \left(\frac{\Delta_i}{\sigma_{\Delta}} \right)^2, \quad (4)$$

$$d_{\text{Yukawa}} = \chi^2(t, b, \tau \text{ or } t, b, \tau, \nu), \quad (5)$$

$$d_{\text{proton}} = \max \left(0, \frac{\tau_{\text{min}}}{\tau_p} - 1 \right)^2, \quad (6)$$

$$d_{\nu} = \chi^2(\Delta m_{21}^2, \Delta m_{31}^2, \theta_{12}, \theta_{23}, \theta_{13}), \quad (7)$$

$$d_{\text{hierarchy}} = \sum_j \left(\frac{\log_{10}(M_{j+1}/M_j) - h_j^*}{\sigma_{h,j}} \right)^2, \quad (8)$$

$$d_{\text{complexity}} = w_T \frac{N_{\text{thresholds}}}{N_{\text{thresholds,max}}} + w_R \frac{N_{\text{irreps}}}{N_{\text{irreps,max}}} + w_F \frac{N_{\text{free}}}{N_{\text{free,max}}}. \quad (9)$$

These terms are intentionally not advertised as fundamental derivations. They are operational defects for testing whether structural ranking adds information beyond conventional GUT fitting.

The hierarchy defect $d_{\text{hierarchy}}$ measures pressure from predeclared scale separations. The scales M_j may include M_Z , an intermediate scale M_I , M_{GUT} , and an optional Planck-adjacent cutoff. The target logarithmic gaps h_j^* and widths $\sigma_{h,j}$ must be fixed before the scan or fitted only on the validation ensemble.

The neutrino defect must be evaluated separately for normal and inverted ordering:

$$d_{\nu}^{\text{NO}} = \chi_{\nu, \text{NO}}^2, \quad d_{\nu}^{\text{IO}} = \chi_{\nu, \text{IO}}^2.$$

The benchmark may report the better ordering only as a model-selection result. It must also report the two orderings separately, so that neutrino ordering is not hidden inside a single post-hoc χ_{ν}^2 .

The threshold prior is defined through σ_{Δ} . In the default dimensionless convention, $|\Delta_i| \leq \sigma_{\Delta}$ is called natural, $\sigma_{\Delta} < |\Delta_i| \leq 3\sigma_{\Delta}$ is threshold-sensitive, and $|\Delta_i| > 3\sigma_{\Delta}$ is called unnatural. The value of σ_{Δ} must be predeclared for each model family because it depends on the assumed heavy spectrum and matching convention.

The complexity defect is normalised rather than counted raw. This prevents bookkeeping choices, such as splitting one threshold parameter into two sub-parameters, from dominating the structural score. A logarithmic alternative,

$$d_{\text{complexity}} = \log(1 + N_{\text{eff}}),$$

should be reported as an ablation when model families have very different field content.

6 Response-Closed Weights

The extended benchmark should not use hand-tuned structural weights as its main result. The allowed hierarchy is:

1. equal-weight baseline;
2. response-closed weights;
3. boundary-aware response weights;
4. ablation tests with one sector removed at a time.

The response-closed form is

$$\lambda = \left(\text{Cov}[d] + \eta \frac{\text{tr}(\text{Cov}[d])}{A} I \right)^{-1} (\langle d \rangle - d^*),$$

where d^* is the target defect vector and η regularises the inverse problem.

Unless explicitly stated otherwise, the main benchmark uses non-negative response weights

$$\lambda_a \geq 0, \quad \sum_a \lambda_a = 1.$$

Unconstrained signed response weights may be reported as a diagnostic ablation, but not as the headline result. This prevents the response closure from rewarding a defect by assigning it a negative weight.

The point is methodological: the weights are not the place where the theory is allowed to hide the fit. If response-closed weights fail to improve over simple baselines, that is a genuine negative result.

7 Extended Master Functional

The next SO(10) benchmark should minimise an extended total loss:

$$\mathcal{L}_{\text{SO}(10)}^{\text{ext}} = \chi_{\text{gauge}}^2{}^{(2\text{loop})} + \chi_{\text{Yukawa}}^2 + \chi_{\nu}^2 + F_{\text{MAAT}}^{\text{multi}} + F_{\text{threshold}} + F_{\text{proton}} + F_{\text{complexity}}.$$

Here $F_{\text{MAAT}}^{\text{multi}}$ is not a single global scalar compression. It is a multi-scale defect-field functional:

$$F_{\text{MAAT}}^{\text{multi}} = F_{\text{mean}} + F_{\text{tail}} + F_{\text{cluster}} + F_{\text{scale}}.$$

The components have different roles:

$$F_{\text{mean}} = \sum_a \lambda_a \langle d_a \rangle, \tag{10}$$

$$F_{\text{tail}} = \sum_a \lambda_a^{\text{tail}} q_{95}(d_a), \tag{11}$$

$$F_{\text{cluster}} = \sum_a \lambda_a^{\text{cl}} \frac{C_{\text{max}}^{(a)}}{N}, \tag{12}$$

$$F_{\text{scale}} = \sum_{a,\ell} \lambda_{a,\ell} d_a^{(\ell)}. \tag{13}$$

This is important for GUT physics because rare but severe defects can dominate viability: proton decay, Landau poles, excessive threshold corrections, and Planck-scale sensitivity can be invisible to a mean score but fatal to a candidate.

The sign convention for F_{cluster} must be fixed before running the benchmark. In this note, $C_{\text{max}}^{(a)}$ denotes the largest connected high-defect cluster in sector a , so F_{cluster} is a penalty. Connected low-defect valleys are not folded into this penalty with the opposite sign. They are reported separately through a valley-coherence observable

$$V_{\text{valley}} = \frac{\text{volume of connected low-defect region}}{\text{total scanned volume}}.$$

This avoids mixing “bad cluster” penalties and “good valley” rewards in one ambiguous term.

8 RG Generator-Stationarity

MAAT v1.5 replaces static balance by stationarity under the relevant generator. In $\text{SO}(10)$, the natural generator is RG flow. Therefore a candidate should not merely fit low-energy values; it should remain structurally controlled under changes in scale.

A simple stationarity support is

$$B_{\text{stat}}^{\text{RG}} = \frac{1}{1 + a \left| \frac{d\chi^2}{d \ln \mu} \right| + b \left| \frac{dF_{\text{MAAT}}}{d \ln \mu} \right| + c \left| \frac{dR_{\text{rob}}}{d \ln \mu} \right|}.$$

This term asks whether the GUT candidate is a stable structural valley under the RG generator, rather than a point-like numerical coincidence at one scale.

9 CCI Diagnostics for GUT Regimes

The Critical Coherence Index can be used as a regime diagnostic rather than as a replacement for likelihoods:

$$\text{CCI}_{\text{GUT}} = \frac{D_{\text{destabilising}}}{\epsilon + D_{\text{stabilising}}}.$$

A useful operational split is

$$D_{\text{destabilising}} = \chi_{\text{gauge}}^2 + d_{\text{threshold}} + d_{\text{proton}} + d_{\nu} + d_{\text{hierarchy}}, \quad (14)$$

$$D_{\text{stabilising}} = H + B + V + R_{\text{rob}} + \Gamma_{\text{Yukawa}} + \Gamma_{\text{vacuum}}. \quad (15)$$

Here

$$\Gamma_{\text{vacuum}} = \frac{1}{1 + d_{\text{vacuum}}}$$

is the support associated with explicit vacuum consistency. A minimal operational defect is

$$d_{\text{vacuum}} = d_{\text{breaking}} + d_{\text{tachyon}} + d_{\text{boundedness}},$$

where d_{breaking} penalises failure to reach the desired low-energy subgroup, d_{tachyon} penalises unwanted unstable scalar modes, and $d_{\text{boundedness}}$ penalises an unbounded scalar potential. If no explicit Higgs spectrum or vacuum analysis is supplied, Γ_{vacuum} must be reported as missing rather than silently set to one in the headline result.

This gives a compact classification:

CCI _{GUT} regime	Interpretation
low	coherent candidate; constraints jointly controlled
medium	critical candidate; threshold-sensitive valley
high	structurally stressed; likely artificial or overfit

Table 2: Proposed GUT regime interpretation. This is a diagnostic label, not a proof of physical realisation.

The low/medium/high CCI thresholds are not free interpretive choices. They must be determined on the validation ensemble. With viability labels, the thresholds should be chosen to optimise a predeclared validation criterion such as balanced accuracy or AUROC. Without labels, use fixed validation quantiles, for example q_{33} and q_{67} , and then freeze those thresholds before evaluating the test ensemble.

10 Benchmark Ladder

The extended programme should be versioned. Each step must produce CSV/JSON outputs and figures, not only prose.

Version	Benchmark	Required result
G0	one-loop gauge baseline	reproduce current failure of precision unification
G1	two-loop gauge running	quantify improvement or failure
G2	GUT thresholds Δ_i	test naturalness of required corrections
G3	intermediate scale M_I	compare one-step vs two-step breaking
Y1	b - τ boundary	reproduce current coherent benchmark
Y2	t - b - τ coupled RG	test third-generation texture stability
Y3	neutrino sector	include angles and mass-squared splittings
Y4	texture families	compare 10_H , 126_H , 120_H patterns
M1	multi-scale MAAT	mean/tail/cluster/scale decomposition
S1	stationarity	RG generator-stationarity under scale flow

Table 3: Proposed MAAT–SO(10) extended benchmark ladder.

11 Model-Family Matrix

Thresholds and beta functions are not universal constants of the word “SO(10)”. They depend on the breaking chain, Higgs irreducible representations, intermediate scales, and neutrino mechanism. The benchmark therefore needs a small predeclared model-family matrix rather than an undefined “SO(10)-like” scan.

Family	Breaking chain	Higgs content proxy	Neutrino mechanism
A	SO(10) $\rightarrow$ SU(5) $\rightarrow$ SM	$45_H + 16_H$	type-I seesaw
B	SO(10) $\rightarrow$ Pati–Salam $\rightarrow$ SM	$210_H + 126_H$	type-I/II seesaw
C	SO(10) $\rightarrow$ left-right $\rightarrow$ SM	$45_H + 126_H$	type-I seesaw
D	SO(10) $\rightarrow$ flipped/intermediate chain $\rightarrow$ SM	minimal effective threshold set	benchmark-dependent

Table 4: Illustrative model-family matrix for the extended benchmark. The entries are operational families for testing threshold and texture sensitivity, not complete model constructions.

Each family should be evaluated under the same predeclared defect definitions, baseline models, and train/validation/test split. A model family may win only if its improvement survives this common protocol.

12 Baselines and Falsification

The extended framework is only useful if it is tested against competitors. The minimum baseline set is:

Baseline	What it tests
χ^2 -only	whether standard fit quality already explains the result
threshold-only	whether MAAT is only hiding threshold tuning
complexity-only	whether simple models already dominate
random weights	whether λ -choice is arbitrary
shuffled defects	whether sector coherence matters
energy/action-only	whether structural terms add anything beyond dynamics
equal-weight MAAT	neutral structural baseline
response-closed MAAT	main non-hand-tuned model
multi-scale MAAT	strongest field-level structural model

Table 5: Required baselines for a credible extended SO(10) benchmark.

Falsification rule. If response-closed or multi-scale MAAT does not improve out-of-sample candidate ranking over χ^2 -only, threshold-only, and complexity-only baselines under predeclared metrics, the SO(10) structural layer should be reported as non-informative for that benchmark.

Out-of-sample protocol. The response closure must be estimated without leaking information from the final test set:

- the training ensemble is used only to estimate response-closed λ_a ;
- the validation ensemble is used to tune regularisation η , CCI regime thresholds, and any normalisation constants;
- the test ensemble is used once for final candidate ranking.

The predeclared primary metrics are:

- Spearman ranking correlation with the final viability score;
- AUROC for binary viable/non-viable candidate classification;
- top- k enrichment of viable candidates among the best-ranked models;
- bootstrap confidence intervals for all reported score gaps.

Secondary diagnostics should include leave-chain-out transfer across model families and the shuffled-defect null gap. Metric switching after observing the test ensemble is not allowed.

13 Interpretation

The extended framework makes the original SO(10) bridge sharper. The gauge benchmark does not yet support a strong MAAT claim: it shows a plausible scale and structural stress, but not precision unification. The Yukawa benchmark is more encouraging: it shows that an SO(10)-motivated b - τ boundary can sit in a high-robustness structural valley with a moderate effective bottom-threshold correction.

The correct next question is therefore not:

Does MAAT prove SO(10)?

but:

Does structural selection add predictive information to SO(10) candidate ranking beyond standard GUT diagnostics?

This is a narrower, testable question. It can be answered negatively, and that is precisely why it is scientifically useful.

14 Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

Current experiment folder:

`experiments/maat_so10_structural_selection/`

Current scripts:

```
python3 maat_so10_gauge_2d_fit.py
python3 maat_so10_yukawa_coupled_rg_v04.py
```

Current outputs:

- `outputs/gauge_2d_summary.json`;
- `outputs/yukawa_v04_summary.json`;
- `outputs/maat_selection_landscape.png`.

Data attribution and licence note. The electroweak-scale inputs use standard reference values for M_Z , gauge couplings, and third-generation fermion masses, cited to the Particle Data Group review [7]. Repository JSON/PNG/PDF artifacts are derived analysis and visualisation outputs only. No endorsement by the Particle Data Group or any external collaboration is implied.

15 Limitations

The extended framework still does not provide:

- a complete $SO(10)$ Higgs sector;
- a selected breaking chain;
- a full two-loop implementation;
- threshold-resolved GUT spectra;
- a realistic neutrino fit;
- proton-decay lifetime predictions;
- a full pole-to- $\overline{MS}$ conversion;
- a UV derivation of the structural supports.

These are not cosmetic omissions. They are exactly the tests that determine whether the MAAT- $SO(10)$ bridge becomes a useful benchmark framework or remains only a phenomenological translation layer.

16 Conclusion

The first MAAT-SO(10) paper established a reproducible phenomenological bridge. This extended note turns that bridge into a test programme.

The current evidence is mixed in a productive way: the one-loop gauge sector shows structural stress and insufficient precision, while the Yukawa sector shows a high-robustness compatibility valley. The proposed extension therefore focuses on the right next tests: threshold-aware gauge running, expanded flavor and neutrino sectors, response-closed weights, multi-scale defect fields, RG stationarity, CCI regime classification, and explicit falsification baselines.

The central claim is deliberately modest:

MAAT does not replace SO(10) dynamics. It supplies a structural ranking layer whose value must be tested against conventional GUT baselines.

References

- [1] H. Georgi and S. L. Glashow, “Unity of All Elementary-Particle Forces,” *Physical Review Letters* **32**, 438–441 (1974).
- [2] H. Fritzsch and P. Minkowski, “Unified Interactions of Leptons and Hadrons,” *Annals of Physics* **93**, 193–266 (1975).
- [3] P. Langacker, “Grand Unified Theories and Proton Decay,” *Physics Reports* **72**, 185–385 (1981).
- [4] R. Slansky, “Group Theory for Unified Model Building,” *Physics Reports* **79**, 1–128 (1981).
- [5] M. E. Machacek and M. T. Vaughn, “Two-loop renormalization group equations in a general quantum field theory. 1. Wave function renormalization,” *Nuclear Physics B* **222**, 83–103 (1983).
- [6] V. D. Barger, M. S. Berger, and P. Ohmann, “Supersymmetric grand unified theories: Two-loop evolution of gauge and Yukawa couplings,” *Physical Review D* **47**, 1093–1113 (1993).
- [7] S. Navas et al. (Particle Data Group), “Review of Particle Physics,” *Physical Review D* **110**, 030001 (2024).
- [8] C. Krieg, “Structural Selection in SO(10)-Motivated Unified Field Theories,” preprint (2026).
- [9] C. Krieg, “The Maturation of MAAT: From Projection Diagnostics to Emergent Robustness Closure,” preprint (2026).
- [10] C. Krieg, “A Maximum-Entropy Derivation of Universal Structural Selection,” preprint (2026).
- [11] C. Krieg, “Multi-Scale Defect-Field Structural Selection,” preprint (2026).
- [12] C. Krieg, “Generator Stationarity and Dynamical Defect-Field Geometry,” preprint (2026).